

MODULE 4A • EPISODE 1

Gravity and the Solar System

From Falling Bodies to Kepler's Ellipses

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Gravity and the Solar System

This episode follows one force from a dropped object to the layout of the Solar System. It begins with Galileo's Law of Falling Bodies and asks why every freely falling object near Earth's surface speeds up at the same steady rate. Newton supplies the answer: his Second Law of Motion and his Law of Universal Gravitation together show that the acceleration due to gravity, g , depends on Earth's mass and radius but not on the mass of the falling object. The panel then follows gravity upward, to the International Space Station and the Moon, uses Newton's Third Law to explain the tides, and shows why g is slightly smaller at the equator than at the poles. The episode closes with the 16th- and 17th-century debate between the geocentric and heliocentric models, Kepler's discovery that orbits are ellipses, and the flat, disk-shaped arrangement of the Solar System that places the planets in the Zodiac.

The Panel

Name	Role
Dr. Edward Thorne	Host; astrophysicist
Dr. Lena Petrova	Guest; exoplanet researcher
Dr. Kenji Tanaka	Guest; microbial biologist with a strong interest in astrobiology

Learning Objectives

After this episode you should be able to:

- ◆ State Galileo's Law of Falling Bodies and describe uniform acceleration near Earth's surface.
- ◆ State Newton's Second Law and use $W = mg$ to relate weight, mass, and the acceleration due to gravity.
- ◆ State Newton's Law of Universal Gravitation and explain how it leads to $g = GM/R^2$.
- ◆ Explain why g does not depend on the mass of a falling object, and how this explains Galileo's Law.
- ◆ Describe how g changes with altitude, and explain why astronauts on the Space Station feel weightless.
- ◆ Apply Newton's Third Law to the Earth and Moon, and explain how the Moon produces tides, including spring and neap tides.
- ◆ Explain the two reasons g is smaller at the equator than at the poles.
- ◆ Compare the geocentric and heliocentric models and describe the contributions of Copernicus and Kepler.
- ◆ Define the visible planets, the Ecliptic Plane, and the Zodiac, and name other objects that orbit the Sun.

QUICK REFERENCE

Key Terms

Term	Definition
Galileo's Law of Falling Bodies	Excluding aerodynamic factors, objects of different mass released from the same height hit the ground at the same time.
Freely falling object	An object falling only under the influence of gravity.
Uniform acceleration	Acceleration that stays constant; speed changes by the same amount every second.
Acceleration due to gravity (g)	The acceleration of a freely falling object, about 9.8 m/s^2 near Earth's surface.
Newton's Second Law of Motion	When a net external force acts on an object, the object accelerates: $F = ma$.
Net force	The total of all the external forces acting on an object.
Weight (W)	The gravitational force on an object; $W = mg$.
Newton's Law of Universal Gravitation	Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers.
Universal gravitational constant (G)	The constant in Newton's Law of Universal Gravitation. Not the same as g.
Free fall	Motion under gravity alone. Astronauts in orbit are in free fall, which is why they feel weightless.
Newton's Third Law of Motion	For every action, there is an equal and opposite reaction.
Tides	The rise and fall of the oceans, caused mainly by the Moon pulling harder on Earth's near side than on its far side.
Spring tides	The largest tidal range, when the Sun, Earth, and Moon are aligned (new and full moon).
Neap tides	The smallest tidal range, when the Sun and Moon are at right angles as seen from Earth (quarter moons).
Effective weight	The weight an object seems to have once the effect of Earth's rotation is included.
Equatorial bulge	Earth's slight flattening; the equator is about 21 km farther from Earth's center than the poles are.
Geocentric model	The model of Ptolemy, with the Earth at the center.
Heliocentric model	The model with the Sun at the center, revived by Copernicus in 1543.
Epicycles	Small extra circles used to make circular-orbit models match the observed motions of the planets.

Term	Definition
Ellipse	An oval curve with two foci. Kepler showed that each planet's orbit is an ellipse with the Sun at one focus.
Visible planets	Planets that can be seen with the naked eye: Mercury, Venus, Mars, Jupiter, and Saturn, plus Earth.
Ecliptic Plane	The plane of Earth's orbit around the Sun, used as the reference plane of the Solar System.
Zodiac	The band of sky about eight degrees north and south of the Ecliptic Plane, through which the Sun, Moon, and planets appear to move.
Other Solar System objects	Asteroids, comets, Kuiper Belt Objects, and the icy bodies of the Oort Cloud, all of which orbit the Sun.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
9.8 m/s²	Acceleration due to gravity near Earth's surface
9.8, 19.6 m/s	Speed of a dropped object after 1 and 2 seconds
~250 mi (~400 km)	Altitude of the International Space Station
~8.7 m/s²	Value of g at the Space Station, about 11% less than at the surface
~2 × 10²⁰ N	Earth's gravitational force on the Moon (and the Moon's on Earth)
9.832 m/s²	Value of g at the poles
9.780 m/s²	Value of g at the equator
9.803 m/s²	Value of g at mid-latitudes such as Chicago
~21 km	How much farther the equator is from Earth's center than the poles are
1543	Publication of Copernicus's On the Revolutions of the Heavenly Spheres
~8°	Width of the Zodiac north and south of the Ecliptic Plane
5 + Earth	The visible planets; Uranus and Neptune generally need a telescope

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Gravity and Falling Bodies

Gravity is the force that shapes the entire Universe, from the orbits of the planets to the structure of galaxies. The episode starts close to home, with **Galileo's Law of Falling Bodies**: excluding aerodynamic factors, objects of different mass released from the same height hit the ground at the same time.

The reason is that **freely falling objects**, those falling only under the influence of gravity, fall with a **constant, or uniform, acceleration** of about **9.8 meters per second, every second**. Starting from rest, a falling object moves at 9.8 m/s after one second, 19.6 m/s after two seconds, and so on.

Part 2 — Newton's Laws of Motion and Gravity

Newton's Second Law of Motion says that when a net external force acts on an object, the object accelerates: $F = ma$. For a freely falling object, the only force is gravity, which we call its **weight**, W , so the Second Law becomes $W = mg$, where g is the acceleration due to gravity.

Weight itself comes from **Newton's Law of Universal Gravitation**: every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between their centers. Your weight is the gravitational force between your mass and Earth's.

Combining the two laws gives $g = GM/R^2$, where G is the universal gravitational constant, M is Earth's mass, and R is Earth's radius. The falling object's own mass cancels out, and M and R are essentially constant near the surface, so every freely falling body there has the same constant acceleration, whatever its mass. This **explains Galileo's Law**.

Part 3 — Gravity Beyond the Surface

Because gravity weakens with distance, g declines as we move away from Earth. At the **International Space Station**, about 250 miles up, g is about **8.7 m/s²**, only about 11% less than at the surface. The astronauts still have plenty of weight; they feel weightless because they and the station are in a constant state of **free fall**.

Even at the distance of the Moon, g isn't zero. Earth pulls on the Moon with a force of about **2×10^{20} newtons**, the force that keeps the Moon in orbit.

Newton's Third Law of Motion says that for every action there is an equal and opposite reaction, so the Moon pulls on Earth with the same force. Because the Moon pulls a little harder on the side of Earth facing it than on the far side, the pull stretches the oceans slightly, producing the **tides**.

The Sun contributes too. When the Sun, Earth, and Moon are aligned, at new moon and full moon, we get the largest tides, the **spring tides**. When the Sun and Moon are at right angles, at the quarter moons, we get the smaller **neap tides**.

Part 4 — Gravity and Latitude

The measured value of g also changes with latitude, mostly because Earth rotates. Every object away from the poles travels in a circle once a day, and moving in a circle takes a small inward force. A little of Earth's gravitational pull is used up providing it, leaving less to press the object against the ground. Objects at the equator make the biggest circles and have the greatest reduction in **effective weight**; objects at the poles have none.

A second effect adds to the first: Earth **bulges slightly at the equator**, so someone standing there is about 21 kilometers farther from Earth's center than someone at a pole, which lowers g a little more.

The differences are small but measurable: g is about **9.832 m/s²** at the poles, **9.780 m/s²** at the equator, and about **9.803 m/s²** at mid-latitudes such as Chicago. Your weight changes slightly depending on where you are on Earth.

Part 5 — The Solar System Models

In the 16th and 17th centuries, astronomers argued over two models: the **geocentric model** of Ptolemy, with the Earth at the center, and the **heliocentric model**, with the Sun at the center. The Sun-centered idea goes back to the ancient Greek astronomer **Aristarchus**, but it was the Polish cleric **Copernicus** who revived it as a complete Solar System model in *On the Revolutions of the Heavenly Spheres*, published in 1543 as he lay dying.

Copernicus was right to put the Sun at the center, but he assumed the planets moved in perfect circles. To match the sky he still needed **epicycles**, the small extra circles Ptolemy had used, so in practice his model was not much simpler or more accurate than Ptolemy's, which slowed its acceptance.

Johannes Kepler, a contemporary of Galileo, fixed the flaw by showing that planetary orbits are **ellipses** with the Sun at one focus. That insight, together with Galileo's telescope observations, led to the general acceptance of the heliocentric model.

The **visible planets** are those seen with the naked eye: Mercury, Venus, Mars, Jupiter, and Saturn, plus Earth itself. Uranus and Neptune generally require a telescope; Uranus is at the edge of naked-eye visibility under very dark skies.

Because the Solar System formed from a thin disk of gas and dust, the planets orbit in roughly the same plane. The plane of Earth's orbit serves as the reference, the **Ecliptic Plane**, and the band about eight degrees north and south of it is the **Zodiac**. The planets' orbits lie within it, which is why the Sun, Moon, and planets move through the zodiacal constellations. Asteroids, comets, Kuiper Belt Objects, and the icy bodies of the Oort Cloud also orbit the Sun.

Part 6 — Conclusion

The episode traces one idea from Galileo's falling bodies to Newton's universal laws, showing that the same gravity that pulls a dropped object to the ground also holds the Moon in orbit and raises the tides. It sets the stage for a closer look at Kepler's revolutionary insights into how the planets move.

TRY IT YOURSELF

Hold a book and a sheet of paper at the same height and drop them together. The book wins. Now crumple the paper into a tight ball and try again. They land together, because crumpling removed most of the air resistance, the aerodynamic factor in Galileo's Law.

Then check the numbers: a stone dropped from rest falls for 4 seconds. How fast is it going? (Answer: $4 \times 9.8 = 39.2$ m/s, ignoring air resistance.)

STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Gravity and Falling Bodies

- 1.** State Galileo's Law of Falling Bodies.

- 2.** What is a freely falling object, and what kind of acceleration does it have?

- 3.** What does it mean to say that g is “9.8 meters per second, every second”?

- 4.** An object is dropped from rest. Ignoring air resistance, how fast is it falling after 1, 2, and 3 seconds?

Part 2 — Newton's Laws of Motion and Gravity

1. State Newton's Second Law of Motion and identify each symbol in $F = ma$.

2. What is weight? Write the equation that relates weight to mass.

3. State Newton's Law of Universal Gravitation.

4. If the distance between the centers of two masses doubles, what happens to the gravitational force between them?

5. Which two laws are combined to find g , and what expression results?

6. Explain why g does not depend on the mass of the falling object, and how this explains Galileo's Law.

Part 3 — Gravity Beyond the Surface

- 1.** What happens to g as you move far above Earth's surface? Why?

- 2.** What is g at the International Space Station, about 250 miles up, and how does it compare with the surface value?

- 3.** If gravity at the Space Station is nearly 90% of its surface value, why do the astronauts feel weightless?

- 4.** Is g zero at the distance of the Moon? How large is Earth's gravitational force on the Moon, and what does it do?

- 5.** State Newton's Third Law of Motion and apply it to the Earth and Moon.

- 6.** Explain how the Moon's gravity produces the tides.

7. What are spring tides and neap tides, and when does each occur?

Part 4 — Gravity and Latitude

1. Why does Earth's rotation reduce the measured value of g ?

2. Where on Earth is the reduction from rotation greatest, and where is it zero? Why?

3. What second effect makes g smaller at the equator than at the poles?

4. Give the approximate value of g at the poles, at the equator, and at mid-latitudes such as Chicago.

5. About what percentage less would you weigh at the equator than at the North Pole?

Part 5 — The Solar System Models

1. Distinguish the geocentric and heliocentric models. Which astronomer is associated with the geocentric model?

2. Who first proposed a Sun-centered system, and who revived it as a complete Solar System model? When, and in what book?

3. What was the flaw in Copernicus's model, and what was its consequence?

4. Who fixed the flaw, and how? What else helped the heliocentric model win acceptance?

5. What are the visible planets? Which planets generally require a telescope?

6. Why do the planets orbit in roughly the same plane? What is the Ecliptic Plane?

7. What is the Zodiac, and why do the Sun, Moon, and planets appear in the zodiacal constellations?

8. Besides the planets, name four kinds of objects that orbit the Sun.

Part 6 — Conclusion

1. The episode says the same gravity that pulls a dropped object to the ground also holds the Moon in orbit. Explain how the episode supports this idea.

2. Kepler's ellipses came decades before Newton's laws. Why does the episode treat Newton and Kepler as parts of one story?

Notes